

JOB DESCRIPTION: SENIOR SOFTWARE ENGINEER

1. Job Overview

- **Position:** Senior Software Engineer
 - **Focus:** Search & Recommendation Systems, Real-time Data Pipelines, Search Relevance Optimization
 - **Department:** Smart Mobility / In-car Navigation
 - **Location:** Ha Noi / Ho Chi Minh City, Vietnam
 - **Level:** Senior
 - **Reports to:** Engineering Manager – Map & Navigation
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2. Job Purpose

We are looking for a Senior Software Engineer specializing in Search to lead the design, development, and optimization of our large-scale Point of Interest (POI) and recommendation systems. In this role, you will bridge the gap between backend engineering, data pipelines, and machine learning to deliver high-performance, low-latency, and highly personalized search experiences. You will be responsible for ensuring our search infrastructure is scalable, cost-efficient, and capable of handling complex query intents in real-time.

3. Key Responsibilities

- **Search Architecture & Query Optimization:** Design and implement dynamic query-building systems supporting full-text search capabilities (fuzzy matching, synonym expansion, misspelling tolerance). Optimize search indexing structures and query strategies to maximize resource efficiency and minimize latency.
- **Model Integration & Personalization:** Train, deploy, and maintain Deep Learning/Machine Learning models for intent understanding and query generation. Apply recommendation and re-ranking algorithms to enhance personalized search relevance and user experience.
- **Data Pipeline Engineering:** Architect and maintain robust, large-scale ETL pipelines for data aggregation. Design near real-time data streaming pipelines to synchronize production databases with search indices.
- **Experimentation & Monitoring:** Implement A/B testing frameworks for search systems, including traffic splitting, experiment logging, and performance analysis. Establish search quality tracking mechanisms and build comprehensive dashboards to monitor system health, search latency, and business metrics.

- **System Performance Tuning:** Optimize multi-worker and multi-threading execution to maximize throughput under high-load conditions. Ensure system architectures dynamically scale to balance cost efficiency and system reliability.
 - **Cross-functional Collaboration:** Work closely with Product, Infrastructure, Data Science, and Backend teams to deliver production-ready, end-to-end search solutions.
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4. Requirements

Must-Have Skills

- Proficient in Python and asynchronous frameworks (e.g., FastAPI).
- Strong experience in designing scalable backend services, microservices, and containerized deployments using Docker and Kubernetes.
- Advanced hands-on experience with **Elasticsearch** (index management, custom analyzers, cluster tuning) and caching solutions like Redis.
- Strong experience in building large-scale batch data aggregation and ETL workflows using **Apache Airflow**.
- Experience with real-time streaming and message brokers (e.g., Google Pub/Sub, Kafka) to handle high-throughput live data synchronization.
- Solid foundation in integrating and serving ML/DL models in production using frameworks like PyTorch and Scikit-learn.
- Familiarity with model serving optimization tools (e.g., ONNX, Triton Inference Server).
- Understanding of search relevance tuning, intent classification, and re-ranking algorithms.

Nice-to-Have

- Experience working with map/routing engines or geospatial search (e.g., Valhalla, OSRM).
- Cloud platform experience, preferably Google Cloud Service (GCS).
- Knowledge of CI/CD pipeline automation (GitLab CI/CD) and automated load testing (K6).

Soft Skills & Mindset

- **Analytical Mindset:** Data-driven approach to solving complex search and ranking problems.
 - **Performance-Oriented:** Obsessed with sub-second latency, computational resource optimization, and cost efficiency.
 - **Collaboration & Leadership:** Excellent communication skills to bridge technical execution with product vision; ability to mentor junior engineers.
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5. Success Metrics

Your performance and impact will be measured by the following targets:

- **Top 1-2-3 Selection Rate:** Improve search accuracy and relevance.
 - **Search Latency:** Optimize search query performance, ensuring low-latency responses during peak traffic.
 - **Abandonment Rate:** Significantly reduce the search abandonment rate by improving fuzzy matching, intent understanding, and synonym expansion.
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6. What We Offer

- **Attractive Salary:** Highly competitive gross salary reflecting your senior expertise, plus a performance-based bonus scheme.
 - **Thirteenth-Month & Performance Salary:** 14 to 16 months of salary per year, depending on business performance and individual impact.
 - **Lunch Allowance:** Fully covered daily lunch allowances.
 - **Learning & Development Fund:** Budget allocated for continuous learning, technical certifications, and attending tech conferences.
 - **Comprehensive Insurance:** Full social insurance (BHXH, BHYT) in compliance with labor laws, plus premium private health care packages.
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7. How to Apply

Interested candidates are invited to submit their CV detailing relevant project achievements to our recruitment team:

- **Email:** [Insert HR Email Here]
- **Subject Format:** [Apply] Senior Search Engineer - [Your Full Name]

Join us to build the next-generation search engine that powers millions of real-time transactions daily!